

Recap:

Monge Problem

Goal: $T: X \rightarrow Y$ that moves some mass μ to a target ν (as cheaply as possible)

$\mu, \nu \rightarrow$ Probability measures

$$\min_T \left\{ \int_X c(x, T(x)) d\mu(x) : T\# \mu = \nu \right\}$$

X (Source space) Y (Target space)

$$(T\# \mu)(A) := \mu(T^{-1}(A)) = \nu(A)$$

Conserving mass (starting mass in X
= capacity in Y)

Kantorovich Problem

$$\min \left\{ \int_{X \times Y} c d\gamma : \gamma \in \Pi(\mu, \nu) \right\}$$

$c \rightarrow$ Cost function

$\gamma \rightarrow$ Transport Plan (Matrix)

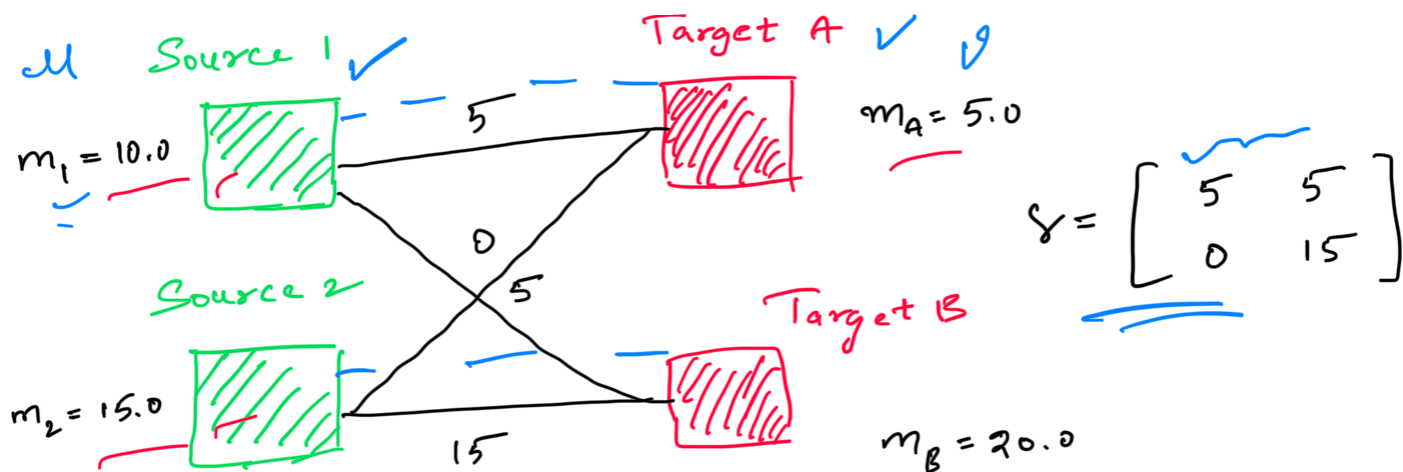
valid transport plans

min transport plans

↳

$$\pi(\mu, \nu) = \{ \gamma \in P(X \times Y) : (\pi_x)_\# \gamma = \mu, (\pi_y)_\# \gamma = \nu \}$$

π_x, π_y are projections of $X \times Y$ onto X, Y resp.



Monge Vs Kantorovich

Duality

$$\min(CP) = \max(DP)$$

$\phi, \psi \rightarrow$ Continuous functions

$$\max \left(\int \phi d\mu + \int \psi d\nu \right) = \min \left(\int c d\gamma \right)$$

Restricted by the constraint

$$\phi + \psi \leq c$$

The cost of two 1D arrays (ϕ, ψ) combined cannot exceed the cost of actual route 2D matrix

Computationally ϕ & ψ are array of prices
 (source distribution mass) (Target distribution mass)

Minimizing the distance
 b/w these distributions

Existence of Optimal T (Monge Map)

$$\underline{c(x, y)} = h(x - y)$$

h $\rightarrow$ strictly convex fn
 $(|x - y|^2)$ ✓

$$\phi(x) + \psi(y) = c(x, y) = h(x - y)$$

($\because$ Optimal Condition)

$$\nabla \underline{\phi(x)} = \nabla h(x - y) \quad (\because \text{first order derivative})$$

h is strictly convex, ∇h is strictly increasing (invertible)
 w.r.t. x

$$x - y = (\nabla h)^{-1}(\nabla \phi(x))$$

$$\underline{y} = \underline{x - (\nabla h)^{-1}(\nabla \phi(x))} =: \underline{T(x)} \quad (\text{Deterministic map})$$

The chapter 3 from computational book
 is rewritten in discrete lang (measures
 become vectors, transport plan & becomes

matrix P and Integrals become Sum)

3.1 The Kantorovich Linear Programs

(Loss Matrix)

$$L_c(a, b) = \min_{P \in U(a, b)} \sum_{i,j} C_{i,j} P_{i,j}$$



(Our Primal OT Problem)

Cheapest possible way to shift histogram A to histogram B

$U(a, b) \rightarrow$ Valid transport plan

A linear thing minimized und linear set of constraints is by definition linear program

$P \rightarrow \underline{n \times m} \xrightarrow{\text{flatten}} \text{vector } \underline{P}$

Rule: rows sum to a , columns sum to b

$\begin{bmatrix} a_i \end{bmatrix}$

$$\underline{A} = \begin{bmatrix} 1_n^T & \otimes & -I_m \\ I_n & \otimes & 1_n \end{bmatrix} = \begin{bmatrix} \{a_i\} \\ \{b_i\} \\ \{b_r\} \end{bmatrix}$$

Computes the flattened row sums & col sums in matrix multiply

$$A_p = \begin{bmatrix} a \\ b \end{bmatrix}$$

A_p says if you add up rows & cols you end up with supply a (source) and demand b (target)

$$L_c(a, b) = \min_{\substack{p \geq 0 \\ A_p = \begin{bmatrix} a \\ b \end{bmatrix}}} c^T p$$

Dual Problem (C-Transforms)

Every LP has a twin problem

$\phi + \psi$

$$L_c(a, b) = \max_{h \in \mathbb{R}^{n+m}} \begin{bmatrix} a \\ b \end{bmatrix}^T h$$

* $A^T h \leq c$

$h \rightarrow$ two arrays stacked (dual potentials)

$f \rightarrow$ price for source bin $\underline{f}$ and $\underline{g}$
 $g \rightarrow$ price for target bin

Duality says,

$f_i + g_j \leq c_{ij}$ (for every route) $\underline{c_{i,j}}$
 (similar to $\phi + \psi \leq c$)

$\min(C_{ij}) = \max(f_i + g_j)$

$L_c(a, b) = \max_{(f, g) \in R(c)} \langle f, a \rangle + \langle g, b \rangle$
 $\max(\int p du + \int \phi dv)$

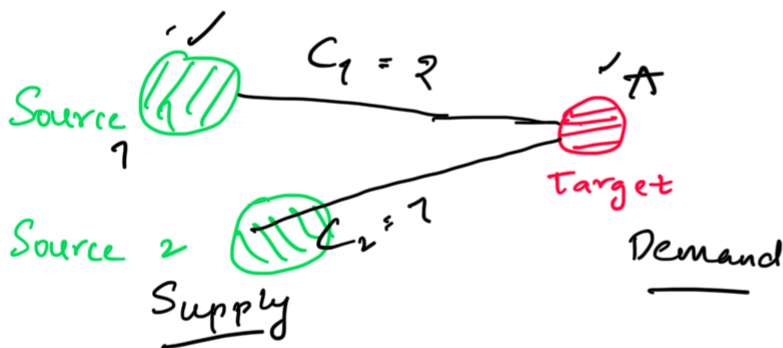
Given f , the C -transform is the tightest feasible g

$(f^c)_j = \min_i (c_{ij} - f_i)$

$L_c(a, b) = \max_{f \in R^+} \langle f, a \rangle + \langle f^c, b \rangle$

(maximizing over single var)

Example



(Given $\underline{f_1} = 1, \underline{f_2} = 3$)

Highest feasible g ?

(at source)

Duality $\rightarrow$ Pick up fee f_i and drop off fee g (at target)
 Constraint: $\underline{f_i + g} \leq C_i \checkmark$

$\rightarrow$ Via Source 1, $1 + g \leq 2, g \leq 1 \checkmark$

$\rightarrow$ Via Source 2, $3 + g \leq 7, g \leq 4 \checkmark$

$$\min(1, 4) \rightarrow 1$$

$$g = f^c = 1 = \underline{\min(C_1 - f_1, C_2 - f_2)}$$

Complementary Slackness

At Optimum, mass only flows on routes where two dual prices add up exactly to the cost

$$f_i + g_i = C_{i,j}$$

If a pair of feasible primal & variables are complementary then we conclude they are optimal

Entropy Regularization

Antenna ... plan

entropy of a plan

$$H(P) = - \sum_{i,j} P_{ij} (\log P_{ij} - 1)$$

$P \rightarrow$ Transport plan
(i to j)

(strongly concave)

$H(P) \rightarrow$ single number on how spread out the plan is (blurriness)

Plug into Original OT

$$\underline{L_c^\epsilon(a,b)} = \min_{P \in \mathcal{U}(a,b)} \langle P, c \rangle - \epsilon H(P)$$

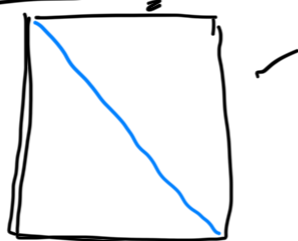
OT cost that controls blur

Idea: Cheapest plan

$\epsilon \rightarrow$ more blurry

Exact OT ($\epsilon \rightarrow 0$)

Sparse matrix

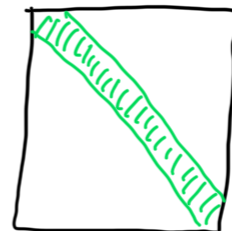


Sharp (fewer routes)

$\downarrow$
Time $O(n^3)$ can't

be used by NW
(non-differentiable)

Entropic OT ($\epsilon > 0$)



blurry (spread out)

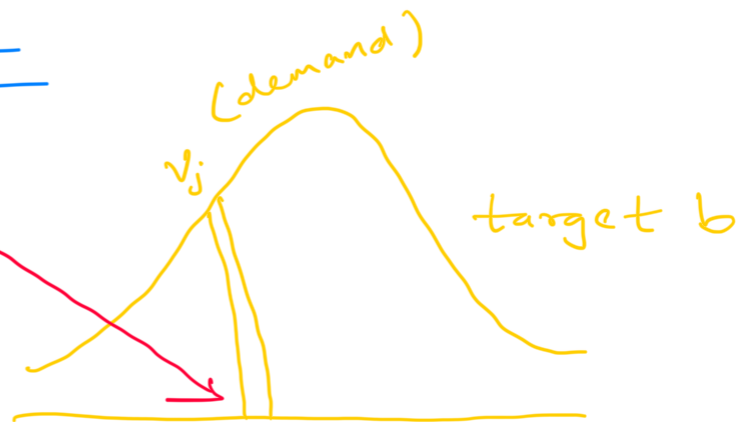
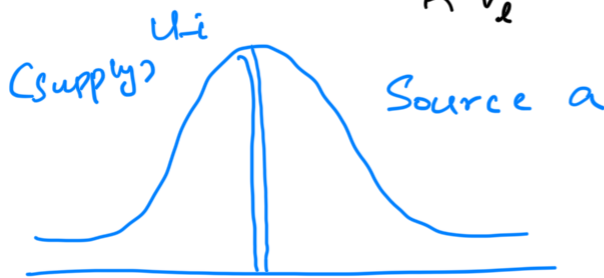
(Convex and unique answer)
can be solved
by Sinkhorn Algorithm

Sinkhorn Phenomenon

$$K = e^{-C/\varepsilon}$$

K is the table that runs Cost matrix $e^{-C/\varepsilon}$ cheaper routes become near 1 and expensive routes near 0.

$$u_{k+1} \leftarrow \frac{a}{K v_k}, \quad v_{k+1} \leftarrow \frac{b}{K^T u_k} \quad (\text{repeat})$$



K_{ij} = closeness

mass on route $i \rightarrow j$

$$= u_i \cdot K_{ij} \cdot v_j$$

$$\hookrightarrow p_{ij} = u_i K_{ij} v_j$$